

13-popquiz capacitors

 This is a preview of the published version of the quiz

Started: Mar 22 at 7:11pm

Quiz Instructions

<https://phet.colorado.edu/sims/cheerpj/capacitor-lab/latest/capacitor-lab.html?simulation=capacitor-lab>  (<https://phet.colorado.edu/sims/cheerpj/capacitor-lab/latest/capacitor-lab.html?simulation=capacitor-lab>)

Select wisely.



Question 1 15 pts

[capacitor APP](https://phet.colorado.edu/sims/cheerpj/capacitor-lab/latest/capacitor-lab.html?simulation=capacitor-lab)  (<https://phet.colorado.edu/sims/cheerpj/capacitor-lab/latest/capacitor-lab.html?simulation=capacitor-lab>)

Open the app in your browser.

- 1) Go to dielectric tab.
- 2) Upper right : check all the boxes.
- 3) Increase the voltage to 1.5V.
- 4) Shove the the dielectric between the plates.

What is true (compare before you shove and after you shove. As you shove:



charges, energy, capacitance increase



charges increases, energy stays constant, capacitance increases



capacitance increases, energy stays constant, charges stay constant



charges stay constant, energy increases, capacitance increases



Question 2 15 pts

Building on the previous question. The dielectric is a non-conducting material. What is true inside the dielectric.



The electric field can penetrate the dielectric and the material get polarized



the electric field can not penetrate the material



the electric field can penetrate the material and there is no polarization inside



there is polarization and there is no electric field inside



Question 3 15 pts

Building on the previous question. Shoving the dielectric between the plates, you (as an external agent) are doing:

hint: positive work is done on a system if the energy of the system increases (the energy of the agent decreases). negative work is done if the system loses energy and the agent gains energy. Work is energy transferred.



positive work



negative work



no work



can't tell



Question 4 15 pts

Building on previous question. Keep the same configuration. You see that the capacitance is 0.44 pF (pico is 10^{-12}). The area is 100 mm^2

Increase area to 400 mm^2 . Look at the new capacitance. What is true (divide the new capacitance by the old capacitance. the pico cancels out)

☐

the capacitance increases by a factor of 4

☐

the capacitance decreases by a factor of 4

☐

the capacitance stays constant

☐

the capacitance increases by a factor of 2

☐

the capacitance increases by a factor of 3



Question 5 15 pts

Building on the previous question. The area is 400mm^2 . the distance between the plate is 10mm decrease the distance to 5mm. What is true (forget about pico , it cancels out). Divide the new capacitance by the old one.

☐

the capacitance increases by a factor of 2

☐

the capacitance decreases by a factor of 2

☐

the capacitance stays the same

☐

the capacitance increases by a factor of 3



Question 6 15 pts

Little girl is isolated from the ground and touches a charged Van Graaf generator . The metallic dome has a potential of 5000V. what is true. No second attempt



- ☐ inside the dome the potential is 5000V and the electric field is 0
- ☐ both the potential and the electric field are 0 inside
- ☐ the potential inside the dome is 0 and the electric field is a constant
- ☐ I didn't pay attention in class.



Question 7 10 pts

building on the previous question. The little girl is isolated from the ground (her shoes are very rubbery) . What is true

- ☐ she is at a potential of 5000V
- ☐ she is at a potential of 0V
- ☐ Her head is at a potential of 5000V and her feet at a potential of 0V
- ☐ I spaced out in class.

Not saved

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